

WILLIAM WIBOWO CIPTONO

Senior AI & Data Engineer

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SUMMARY

Google Cloud Professional Data Engineer (PDE) Certified, with 4+ years of experience in designing, developing, and optimizing large-scale distributed data pipelines and systems. Proficient in Python, SQL, DBT, and Airflow. Master in data ingestion and data modeling, ensuring your data warehouse stored your data in the most *efficient & best-practice-oriented* manner. Developed a standardized third-party data integration framework, aggregating over 30+ data sources while improving ingestion performance through 75% reduction in Kubernetes memory overhead and a whopping 80% decrease in cost. Contributed in the development and design of an end-to-end pricing strategy ETL pipeline, which directly contributes up to ~10% revenue increase.

EXPERIENCE

Gojek – Jakarta, Indonesia

Senior Data Warehouse Engineer (September 2025 – Present)

- Nurtured 5 engineer team members to improve data quality's timeliness score (SLA delivery met) from 93% to 99.5% within 3 months through house keeping, SOP alignment, and report + alert enablement.
- Analyzed more than 1000 data ETL pipelines and implemented schedule preponement that accelerated the P99 finished task from 12PM UTC+7 to 9AM UTC+7.
- Designed & implemented cloud cost optimization strategy, which resulted in saving 60% data warehouse (Alibaba MaxCompute) monthly budget.
- Developed several end-to-end reverse ETL pipelines to dump data into both internal & external systems, including but not limited to: AliCloud OSS, SFTP, MySQL, internal segmentation service, etc.

Traveloka – Tangerang, Indonesia

Senior Data Engineer - Data Warehouse Engineer (March 2025 – September 2025)

- Led a team of 2 members within the Commercial Department Assets Migration and Remodel project. Migrating and standardizing more than 500 assets from BigQuery Scheduled Query and Views to DBT Airflow. Created a tracking dashboard in Looker Studio to track the migration progress and improve visibility towards the stakeholders.
- Developed DBT Source Freshness Check and Tableau Data Source Auto Refresh Connector written in Python to automatically refresh tableau data source when the data is ready. Removed 100% operational effort in maintaining the tableau data source reliability. Improved overall dashboard timeliness from 70% to 90% within a single month.

Traveloka – Tangerang, Indonesia

Data Engineer - Data Warehouse Engineer (October 2023 – March 2025)

- Developed a standardized third-party data ingestion (external data ingestion) framework written in Python, aggregating 30+ data sources, improving ingestion code maintainability and development process. Significantly optimized data ingestion processes, achieved a 75% reduction in Kubernetes memory overhead, a 50% improvement in ingestion speed, and an 80% decrease in annual third-party data ingestion costs across Traveloka.
- Developed price stance & dynamic pricing point pipeline using DBT and BigQuery for flight and accommodation products to empower Traveloka's pricing strategy team. Contributed to ~10% yoy revenue increase within less than 1 year.
- Enabled LLM-powered customer service data pipeline model using Retrieval Augmented
- Graph (RAG) pattern to generate flight reschedule recommendation for customer service team using Vertex AI.
- Maintained data quality and governance through custom alerts, data remodeling, and BigQuery slot optimization, improving Traveloka's data delivery timeliness from 60% to 98% within 4 months.
- Initiatively conducted 4 training & coaching programs related Data Warehousing and Data Modeling concepts, targeted for all members of Traveloka's data team.

Shopee Ltd – Jakarta, Indonesia

Software Quality Assurance Automation Engineer - Marketplace (April 2023 – October 2023)

- Developed more than 200 automation scripts for Shopee's iOS App using the XCUITest framework written in Objective C programming language. Specialized in the user marketplace logistics (MPL), marketplace orders, and user order lists features.
- Developed more than 10 automation scripts for Shopee's Android App using the UiAutomator and Espresso frameworks in Java. Specialized in user identification features.

- Led a team of 20 QA testers in final live regression testing for Marketplace Logistics (MPL), New Payment Channel Integration (MPP), User Cart, and Seller Wallet features, ensuring app reliability by executing over 250 P0 regression test cases weekly.

Shopee Ltd – Jakarta, Indonesia

Junior Full Stack Software Engineer - Digitalent (September 2022 – April 2023)

- Led a team of 5 developers to create a prototype e-commerce app within 2 months, utilizing Golang for the backend and React for the frontend. Utilizing Typescript, OAuth, Redis Caching, CI/CD Pipeline, and Cloud Deployment.
- Developed prototype e-wallet and restaurant booking apps using Golang for the backend and React for the frontend.
- Developed and organized a DigiTalk talk show, showcasing SeaTalk (internal company chat platform) chatbot development, attended by over 100 participants.

PT VISIONET INTERNASIONAL (OVO) – Jakarta, Indonesia

Business Intelligence - Data Platform Engineer (December 2021 – September 2022)

- Assessed, designed and implemented the end-to-end data pipeline migration process from Spark, Hive, and Impala on-premise server into the new Google Cloud Platform (GCP) BigQuery.
- Developed Cross-Platform Task Dependency feature on Airflow using REST API to streamline job scheduling BAU. Eliminating 100% manual effort in maintaining dependencies between platform and DAG.
- Developed automatic report Email Blast Airflow Operator written in Python. Enabled more than 20 automatic email reports.
- Conducted User Requirement Analysis, Data Modelling, ETL development (BigQuery, Hive, Impala), dashboard development (Tableau, Google Data Studio), and job scheduling management (Airflow).

Paragon Technology And Innovation – Jakarta, Indonesia

Master Data Management Intern (July 2021 – September 2021)

- Migrated and standardized over 50,000 B2B and B2C customer records from more than 30 data points with different formats to the new standardized ERP system. Developed data preparation and migration procedures to ensure the data integrity.
- Developed a custom data preparation tool written in VBA (Visual Basic for Application). Eliminated data preparation time by up to 75% and significantly reduced the risk of human error.
- Migrated and standardized more than 15,000 product production routing data spread across 4 different factories.

EDUCATION

INSTITUT TEKNOLOGI BANDUNG (2018 - 2022)

Industrial Engineering

Wrote a thesis related to Natural Language Processing, “Develop Topic Modelling Tool for Extracting Topic from PeduliLindungi Apps Reviews Using Text Mining Method” in collaboration with the Informational & Decision System Expertise Group ITB (KKSIIK ITB).

CERTIFICATION

GOOGLE CLOUD (2024 - 2026)

Professional Data Engineer (PDE)

A Professional Data Engineer makes data usable and valuable for others by collecting, transforming, and publishing data. This individual evaluates and selects products and services to meet business and regulatory requirements. A Professional Data Engineer creates and manages robust data processing systems. This includes the ability to design, build, deploy, monitor, maintain, and secure data processing workloads.

SKILLS

- Programming Language: Python, Go, Javascript, Bash Script
- Query: SQL (BigQuery, PostgreSQL, Impala SQL, Hive SQL, MsSQL, MySQL, SQLite, etc.)
- Data Concepts: Medallion Architecture, Data Modelling, Kimball and Inmon, Star Schema & Snowflake Schema, Data Vault, Semantic Layers, ETL vs ELT, Data Ingestion, Reverse ETL, Data Governance, OLAP vs OLTP.
- Data Pipeline Tool: DBT (Data Built Tool), Airflow
- Data Visualization tool: Tableau, Looker Studio, Python matplotlib
- Infrastructure: Containerization (Docker), Orchestration (Kubernetes), and Infrastructure as a Code (Terraform)
- Cloud: Google Cloud Platform (GCP), Amazon Web Service (AWS), Alibaba Cloud
- Algorithm and Data Structure
- Development Architecture: Clean Architecture, Atomic Design
- Web Development: React, NextJS, NodeJS, Handlebar, Gin